

An atomic isotropic counterexample to the log-free ratio conjecture

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Source conjecture. For a probability measure μ on $\mathbb{R}^d$, let μ_m be the empirical measure of m independent samples. Bartl and Mendelson [1] consider

$$\mathcal{U} = \left\{ \{x : \langle \theta, x \rangle \in I\} : \theta \in \mathbb{R}^d, I \subset \mathbb{R} \text{ a generalized interval} \right\}.$$

Their Theorem 2.4 gives absolute $c_0, c_1 > 0$ such that, if $m \geq c_0 d / \Delta$, then with probability at least $1 - \exp(-c_1 \Delta m)$,

$$|\mu_m(A) - \mu(A)| \leq \sqrt{\Delta \mu(A)} \log \left(\frac{e}{\mu(A)} \right) \quad (1)$$

simultaneously for $A \in \mathcal{U}$ with $\mu(A) \geq \Delta$. Remark 2.5 conjectures that the logarithmic factor can be removed.

2.2 A generalization of the DKW inequality for linear functions

A key component in the proof of Theorem 1.4 is the following uniform estimate on the deviation between the empirical and the true distribution functions for each marginal μ^θ . Let

$$\mathcal{U} = \left\{ \{ \langle \theta, \cdot \rangle \in I \} : \theta \in \mathbb{R}^d, I \subset \mathbb{R} \text{ is a generalized interval} \right\},$$

where by “generalized interval” we mean an open/closed, half-open/closed interval in $\mathbb{R}$, including rays.

Theorem 2.4. *There are absolute constants c_0, c_1 such that the following holds. Let $\Delta \leq 1$ and $m \geq c_0 \frac{d}{\Delta}$. Then, with probability at least $1 - \exp(-c_1 \Delta m)$, for every $A \in \mathcal{U}$ satisfying $\mu(A) \geq \Delta$, we have that*

$$|\mu_m(A) - \mu(A)| \leq \sqrt{\Delta \mu(A)} \log \left(\frac{e}{\mu(A)} \right).$$

Remark 2.5. A version of Theorem 2.4 was proved in [26], with an error estimate of the order $\sqrt{\Delta \mu(A)}$ (without a logarithmic factor) but with the restriction that $\Delta \geq c \frac{d}{m} \log(\frac{em}{d})$. That restriction results in a suboptimal estimate in Theorem 1.4 while the optimal one, at least for $q > 4$, can be derived from Theorem 2.4. We conjecture that the logarithmic factor in Theorem 2.4 can be removed, but its removal does not affect the estimate in Theorem 1.4 for $q > 4$, nor will it eliminate the logarithmic factor when $q = 4$.

The proof of Theorem 2.4 is a simple outcome of Talagrand’s concentration inequality for classes of uniformly bounded functions [35], applied to a class of binary valued functions

Here is a counterexample to that conjecture in the stated generality.

Theorem 1. *There do not exist absolute constants $a, b, C > 0$ such that, for every probability measure μ on $\mathbb{R}^d$, every $0 < \Delta \leq 1$, and every $m \geq ad/\Delta$, one has with probability at least $1 - \exp(-b\Delta m)$ the simultaneous log-free estimate*

$$|\mu_m(A) - \mu(A)| \leq C\sqrt{\Delta\mu(A)} \quad (A \in \mathcal{U}, \mu(A) \geq \Delta). \quad (2)$$

The failure already occurs for centered isotropic measures on $\mathbb{R}^2$.

Proof. Assume that a, b, C exist. Fix a constant $\alpha > 2a$ and, for every $N \geq 3$, put

$$v_j = \sqrt{2}(\cos(2\pi j/N), \sin(2\pi j/N)), \quad \mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{v_j}. \quad (3)$$

The roots-of-unity identities give $N^{-1} \sum_j v_j = 0$ and $N^{-1} \sum_j v_j v_j^\top = I_2$; thus μ_N is centered and isotropic. Moreover $|\langle v_j, \theta \rangle| \leq \sqrt{2}$ for $\theta \in S^1$, while isotropicity gives $\|\langle \cdot, \theta \rangle\|_{L_2(\mu_N)} = 1$. Hence these measures also satisfy the paper's running L_q - L_2 marginal equivalence, uniformly in N (indeed with constant $\sqrt{2}$ for every $q \geq 2$).

Every singleton $\{v_j\}$ belongs to $\mathcal{U}$. Indeed, the functional $x \mapsto \langle v_j, x \rangle$ has a strict maximum on the support of μ_N at v_j . Choosing a ray whose endpoint lies strictly between that maximum and the second-largest value isolates v_j .

Set $\Delta_N = 1/N$ and $m_N = \lceil \alpha N \rceil$. For all large N , $m_N \geq 2aN = ad/\Delta_N$. Let K_j be the number of the m_N samples equal to v_j . Applying (2) to every singleton gives

$$\left| \frac{K_j}{m_N} - \frac{1}{N} \right| \leq \frac{C}{N}, \quad 1 \leq j \leq N. \quad (4)$$

Consequently the event in (2) is contained in

$$\left\{ \max_{1 \leq j \leq N} K_j \leq B \right\}, \quad B := \lceil (\alpha + 1)(C + 1) \rceil, \quad (5)$$

where B is independent of N .

We now prove that the probability in (5) tends to zero. On one probability space, generate an infinite sequence of independent uniform choices from $\{1, \dots, N\}$ and an independent random variable $L_N \sim \text{Poisson}(\alpha N/2)$. Let Y_j count label j among the first L_N choices. By Poisson splitting, $Y_1, \dots, Y_N$ are independent $\text{Poisson}(\alpha/2)$ random variables. Hence, with

$$q_B = \mathbb{P}\{\text{Poisson}(\alpha/2) \leq B\} < 1,$$

we have

$$\mathbb{P}\left\{ \max_j Y_j \leq B \right\} = q_B^N \rightarrow 0. \quad (6)$$

Also $\mathbb{P}\{L_N > m_N\} \rightarrow 0$, for example by a standard Poisson Chernoff bound, since m_N is asymptotically twice the mean of L_N . On $\{L_N \leq m_N\}$ the first L_N samples are contained among the first m_N samples, so $Y_j \leq K_j$ for all j . Therefore

$$\mathbb{P}\left\{ \max_j K_j \leq B \right\} \leq q_B^N + \mathbb{P}\{L_N > m_N\} \rightarrow 0. \quad (7)$$

In contrast, (2) asserts that its event has probability at least

$$1 - \exp(-b\Delta_N m_N) \rightarrow 1 - \exp(-b\alpha) > 0.$$

This contradicts (5)–(7), and proves the theorem. $\square$

Remark 1 (What the example does and does not show). *The obstruction is the occupancy fluctuation of small atoms. Thus the logarithmic factor in (1) cannot simply be deleted for arbitrary measures, even if centered isotropic measures are required. The argument does not exclude a log-free theorem under nonatomicity, a uniform density condition, log-concavity, or another regularity assumption. Standard maximum-load asymptotics in fact give $\max_j K_j \asymp \log N / \log \log N$ here, showing that some diverging loss is unavoidable as $\Delta = 1/N \downarrow 0$.*

Status and search. This is a full negative answer to Remark 2.5 as literally stated. The January 2025 source revision still contains the conjecture. Cheap run indexes and bounded exact-title, exact-phrase, and author/topic searches found no prior explicit resolution. The proof is self-contained and uses no numerical computation.

References

- [1] D. Bartl and S. Mendelson, *Structure preservation via the Wasserstein distance*, arXiv:2209.07058, current source revision January 2025.